frontiers in Education

TYPE Original Research
PUBLISHED 06 March 2026
DOI 10.3389/feduc.2026.1749909

Check for updates

OPEN ACCESS

EDITED BY
Semra Demir Başaran,
Erciyes University, Türkiye

REVIEWED BY
Oktay Bektaş,
Erciyes University, Türkiye
Fazilet Özge Maviş Sevim,
Gaziosmanpaşa University, Türkiye

*CORRESPONDENCE
Saltanat Adikanova
adikanovas@gmail.com

RECEIVED 19 November 2025
REVISED 10 February 2026
ACCEPTED 18 February 2026
PUBLISHED 06 March 2026

CITATION
Baltabayeva N, Adikanova S, Ponkina EV and Nurkanova R (2026) Enhancing secondary education through gamification and Bloom's taxonomy: the BilimQuest portal. *Front. Educ.* 11:1749909. doi: 10.3389/feduc.2026.1749909

COPYRIGHT
© 2026 Baltabayeva, Adikanova, Ponkina and Nurkanova. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Enhancing secondary education through gamification and Bloom's taxonomy: the BilimQuest portal

Nazym Baltabayeva[1], Saltanat Adikanova[1]*, Elena V. Ponkina[1] and Raikhan Nurkanova[2]

[1]Sarsen Amanzholov East Kazakhstan University, Ust-Kamenogorsk, Kazakhstan, [2]Kazakh Academy of Sport and Tourism, Almaty, Kazakhstan

This study explores the scientific and methodological foundations for integrating gamification and Bloom's taxonomy into secondary computer science education through the development and testing of the BilimQuest e-learning portal. The portal was designed to structure learning tasks according to Bloom's cognitive hierarchy while embedding motivational game elements such as points, badges, and challenges. An experimental study with 312 fifth-grade students from three schools in East Kazakhstan tested whether integrating gamification with Bloom's taxonomy improves motivation, conceptual understanding, and practical skills. Students were divided into an experimental group using BilimQuest and a control group with traditional instruction. The experimental group used BilimQuest during informatics lessons, while the control group followed traditional instruction. The hypothesis was tested using Pearson's chi-square (χ^2) analysis. Results showed significant improvements across all three components. Teachers noted that the platform's analytics supported real-time progress tracking and informed instruction. Based on these findings, it is recommended that gamified digital learning environments be systematically aligned with Bloom's taxonomy to support higher-order cognitive development and formative assessment in secondary education. The study concludes that the deliberate alignment of gamification with Bloom's taxonomy constitutes a robust pedagogical approach for digital learning environments, with both theoretical and practical relevance for digital pedagogy and the professional preparation of future educators.

KEYWORDS
Bloom's taxonomy, computer science teaching, e-learning portal, gamification, student motivation

1 Introduction

Gamification is increasingly recognized in education as the incorporation of game design elements into learning activities with the aim of strengthening student motivation and engagement. Through elements such as points, badges, levels, and feedback, gamification makes the learning process more interactive and engaging. Research shows that gamified learning environments enhance intrinsic motivation and are widely regarded as effective in fostering participation and social interaction (Sailer and Homner, 2020; Singh and Singh, 2025). For instance, gamification has been found to improve students' attitudes toward learning and encourage active participation, although its impact on academic performance is less consistent. In secondary education, where maintaining student interest is difficult, gamification is increasingly viewed as an effective way to improve lesson engagement (Shaukat et al., 2025). Systematic

reviews also confirm that game elements enhance school engagement when applied within a coherent pedagogical framework (Davletova et al., 2025; Nawaz et al., 2025).

However, existing studies indicate that increased engagement does not necessarily lead to the development of higher-order thinking skills, such as analysis, evaluation, and creation, which are essential for meaningful learning in computer science education. As a result, many students remain focused on surface-level task completion rather than deep cognitive processing and problem-solving.

A key theoretical foundation underlying this work is Bloom's taxonomy of educational objectives. Bloom's framework organizes cognitive goals hierarchically from lower-order (knowledge, comprehension) to higher-order (analysis, synthesis, evaluation, creation) levels (Forehand, 2010). Well-designed learning activities that align with Bloom's levels are thought to promote deep learning and critical thinking. Nevertheless, empirical evidence suggests that secondary school students often experience difficulties in progressing beyond lower-order cognitive levels, particularly in subjects requiring abstract reasoning and algorithmic thinking, such as computer science (Dong et al., 2025; Ritter and Schlomske-Bodenstein, 2025). In game-based learning, aligning tasks to Bloom's cognitive stages can help ensure that games promote not just recall, but higher-order reasoning and problem solving (Plump and LaRosa, 2017). For example, adapting game mechanics to the "apply" level involves strategic challenges, while the "evaluate" and "create" levels involve generating new content and offering feedback. Thus, Bloom's taxonomy provides a pedagogically grounded framework for addressing the observed gap between student engagement and the development of higher-order cognitive skills (Damayanti et al., 2020; Erkek, 2022).

Although gamification and Bloom's taxonomy are each widely recognized as effective approaches, their integrated application in secondary-level instructional design remains limited. To our knowledge, this is one of the first empirical studies to systematically integrate gamification with Bloom's taxonomy within a secondary school computer science curriculum. Most existing gamified learning environments prioritize behavioral engagement and competition, while offering limited scaffolding for cognitive progression across Bloom's levels. As in many countries globally, educational policy in Kazakhstan increasingly prioritizes digitalization and 21st-century competencies such as critical thinking and problem-solving. In particular, national strategic frameworks of the Republic of Kazakhstan, including the State Program for the Development of Education and Science and the "Digital Kazakhstan" initiative, emphasize the transition toward competency-based education and the effective use of digital technologies in school curricula, thereby underscoring the need for pedagogically grounded digital tools that support these priorities in classroom practice (Ministry of Education and Science of the Republic of Kazakhstan, 2020).

To address this gap, the present study develops and tests an interdisciplinary e-learning portal—BilimQuest—that embeds gamified learning tasks systematically organized according to Bloom's taxonomy. By combining educational theory, digital innovation, and psychological insights into learner motivation, BilimQuest is designed as an 'innovation gateway' to modernize curriculum delivery and enhance cognitive engagement in secondary computer science education. The portal is intended not only to increase student motivation but also to support the gradual development of problem-solving and critical thinking skills through cognitively sequenced learning activities. Accordingly, this study has three main objectives:

(1) To analyze the theoretical and methodological foundations of integrating gamification and Bloom's taxonomy in education;
(2) To establish the scientific and methodological framework for incorporating BilimQuest into secondary computer science education;
(3) To empirically evaluate its impact on students' learning outcomes, motivation, and engagement.

Based on the identified challenges in students' motivation and higher-order cognitive development, the study addresses the following research questions:

> RQ1: Does the integration of gamification aligned with Bloom's taxonomy significantly influence students' motivational, cognitive, and practical learning outcomes in secondary computer science education?
>
> RQ2: How does the use of the BilimQuest e-learning portal support teachers' instructional practices and formative assessment?
>
> RQ3: In what ways does qualitative data, treated as secondary data, support and explain quantitative data, considered as primary data, within an embedded mixed-methods research framework?

It was hypothesized that the integration of gamification and Bloom's taxonomy through the BilimQuest e-learning portal would significantly improve students' knowledge levels across motivational, content, and practical components. Specifically, students in the experimental groups using BilimQuest would demonstrate higher motivation, deeper cognitive engagement, and better learning outcomes compared to those taught through traditional methods.

Accordingly, the following hypotheses were formulated:

> H_0: There is no statistically significant difference between the experimental and control groups in terms of motivational, content, and practical learning components.
>
> H_1: There is a statistically significant difference between the experimental and control groups in terms of motivational, content, and practical learning components.

2 Literature review

2.1 ICT tools for gamified learning

More studies have, in turn, been aimed at how information and communication technologies (ICT) can be used to bring about gamification in the learning process. Researchers have experimented with quiz sites, programming interfaces, virtual digital games, and learning management systems with gamified modules. Most of the tools were meant to motivate and engage students and, in some cases, enhance academic

performance. Nevertheless, their core function more often than not is real-time engagement and feedback, not taxing intellectual acquisition.

Among quiz-based systems, Kahoot! and Quizizz became the most employed tools in schools and universities. They have gained attention for increasing student attention, motivation, and classroom participation, cultivating a competitive yet playful learning atmosphere (Boulden et al., 2021; Licorish et al., 2018; Sailer and Homner, 2020; Wang and Tahir, 2020). These tools tend to aim at recognition and recall tasks, thus targeting only the lower levels of cognitive activity (Dicheva et al., 2015). Consequently, they enhance participation but do not guarantee deeper understanding or critical thinking skills development.

Other formative assessment tools, like Socrative and Gimkit, give real-time feedback to the teacher and have simple analytics available for teacher use (Boud and Falchikov, 2007; Nebel et al., 2016). Such systems are effective in keeping track of leaner progress but provide only limited scaffolding for higher-order thinking. Immersive environments, such as Minecraft: Education Edition, have also seen increased use in both STEM and coding education. They foster creativity, collaboration, and problem-solving (Hamari et al., 2016; Ke, 2016; Southgate et al., 2020). However, their open-ended nature requires strong teacher guidance to connect gameplay with curricular goals.

Similarly, coding environments such as CodeCombat, Code.org, Scratch, and Tynker are powerful tools for introducing programming and computational thinking (Fernandes-Henriques and Carvalho, 2021; Grover and Pea, 2013; Resnick et al., 2017). They help students persist through challenges and develop creative approaches to coding but rarely include structured links between learning tasks and specific cognitive stages. Broader gamified ecosystems such as Classcraft and learning management systems like Moodle or Canvas with gamification plug-ins have shown improvements in participation, teamwork, and course completion (Dichev and Dicheva, 2017; Stoecklin, 2020; Sailer et al., 2017). Yet these systems primarily reward behavioral activity-such as logging in or completing modules-rather than measuring conceptual understanding or analytical reasoning.

More recently, H5P gamified modules and adaptive mathematics platforms like Knewton and ALEKS have offered automated feedback and progress visualization (Krathwohl, 2002; Xu and Wang, 2021). Although these technologies provide valuable data about student activity and adaptation to mistakes, their algorithms are not explicitly mapped to Bloom's taxonomy. They adjust to user performance but fail to capture progression through cognitive levels such as application, analysis, and creation.

Taken together, the literature makes one conclusion evident: despite the growing popularity of gamified ICT tools, none of the existing platforms explicitly align their design or analytics with Bloom's taxonomy. Current systems successfully motivate learners and enhance participation but tend to remain behavior-oriented, lacking mechanisms that assess or guide higher-order cognitive development. This gap underscores the need for interdisciplinary solutions that combine pedagogical theory, motivational psychology, and ICT engineering. The BilimQuest e-learning portal proposed in this study addresses precisely this challenge by integrating game elements with structured pedagogical scaffolding grounded in Bloom's taxonomy.

2.2 Bloom's taxonomy in gamified learning

A second line of research addresses the use of Bloom's taxonomy as a framework for organizing learning objectives in gamified contexts. Scholars emphasize that gamification should not only capture attention but also guide learners from lower-order skills, such as remembering and understanding, toward higher-order skills, including analyzing, evaluating, and creating (Krathwohl, 2002).

Kgosietsile (2023) stressed that game-based assessments are most effective when mapped across Bloom's stages, ensuring that learners progress through increasingly complex cognitive tasks. Zainuddin et al. (2023) proposed a model in which badges and scores were explicitly connected to Bloom's levels to scaffold student learning (Sailer et al., 2017). Similarly, Ramírez et al. (2024) reviewed school engagement literature and concluded that gamification can operationalize Bloom's emphasis on active participation and cognitive development.

Despite these advances, the integration of Bloom's taxonomy into gamified learning remains limited. Most tools provide only basic feedback and do not evaluate whether learners are advancing from lower- to higher-order thinking. This reveals a clear methodological gap: while Bloom's taxonomy is well established in educational design, its systematic application in gamified environments is rare (Anderson and Krathwohl, 2001; Sailer and Homner, 2020).

The present study addresses this challenge by developing BilimQuest, an interdisciplinary e-learning portal that explicitly organizes gamified tasks according to Bloom's taxonomy. The system is designed not only to enhance student motivation and engagement but also to provide teachers with structured analytics on cognitive development. In doing so, BilimQuest contributes to the scientific and methodological foundations of integrating gamification into secondary computer science education and enables empirical evaluation of its impact on students' learning outcomes, motivation, and engagement.

3 Methodology

3.1 Research design

This study employed a mixed-method design combining theoretical analysis, pedagogical modeling, and a school-based experiment. Specifically, an embedded mixed-methods design was adopted, in which quantitative data served as the primary data strand to evaluate the effectiveness of the BilimQuest e-learning portal, while qualitative data functioned as a secondary data strand to support and explain the quantitative results by providing contextual insights into the implementation process and teacher experiences. The aim was to evaluate how the BilimQuest e-learning portal-structured according to Bloom's taxonomy and enriched with gamified tasks-supports student learning and teacher readiness in secondary computer science education.

3.2 Research sites and participants

The empirical phase was conducted during the 2024–2025 academic year (September 2024 – May 2025) in three secondary schools in Oskemen, East Kazakhstan Region. For ethical reasons, the participating schools are anonymized and referred to as School A, School B, and School C. A detailed overview of the participant distribution across schools, experimental and control groups is presented in Table 1.

TABLE 1 Distribution of participants across schools and groups.

No	School	Total students	Experimental group (EG)	Control group (CG)
1	School A	104 (46 boys, 58 girls)	52	52
2	School B	128 (58 boys, 70 girls)	64	64
3	School C	80 (40 boys, 40 girls)	40	40
Total	—	312	156	156

A cluster sampling strategy was applied, whereby intact classes were assigned to either the experimental or control group to preserve the natural classroom setting. These students had studied digital literacy in grade 4 using traditional methods and then used BilimQuest in grade 5. In addition, pre-service computer science teachers ($N = 4$) were involved in pilot implementation and training activities. These participants contributed qualitative data through reflective feedback and observations related to the use of the BilimQuest portal. The inclusion of both general and adapted schools created a multi-level environment for assessing the flexibility and inclusivity of the portal.

3.3 Phases of the study

The methodology was implemented in five sequential phases:

(1) Theoretical Analysis. A review of scientific literature and methodological guidelines was conducted to identify existing approaches to gamification, Bloom's taxonomy, and ICT-based learning tools.
(2) Portal Design. The BilimQuest portal was developed to integrate pedagogical principles with technical architecture. The platform used Flask for backend development, HTML/CSS/JavaScript for the interface, and Railway hosting for stable operation.
(3) Curriculum Alignment. Gamified tasks were linked directly to the grade 5 computer science curriculum. Each activity was mapped to Bloom's taxonomy levels to ensure progression from knowledge recall to creation.
(4) Pedagogical Experiment. The portal was piloted in classroom practice from September 2024 to May 2025. The experimental group used BilimQuest during selected computer science modules, while the control group followed the standard curriculum using traditional instructional methods without gamified digital tools.
(5) Data Collection and Analysis. Student progress was tracked through built-in analytics, including dashboards and diagrams by Bloom's levels. Entry and exit assessments measured learning outcomes, while questionnaires and semi-structured interviews captured teacher and student perceptions.

3.4 Data collection instruments and analysis

Three components were assessed: motivational, content (cognitive), and practical. The motivational component was measured through structured questionnaires focusing on engagement and persistence in learning tasks. The content component assessed students' mastery of computer science concepts aligned with Bloom's taxonomy. The practical component evaluated students' ability to apply knowledge independently in solving tasks and completing assignments within the portal.

To analyze differences between the experimental and control groups, Pearson's chi-square (χ^2) test was applied. This test was selected because the data were categorical and classified into three levels (low, medium, high). The assumptions of the χ^2 test, including independence of observations and sufficient expected frequencies, were met. Descriptive statistics were used to summarize distributions, and comparative analysis was conducted to evaluate group differences.

3.5 Teacher training and professional development

A parallel strand addressed teacher readiness. Pre-service computer science teachers participated in a series of workshops (3×90 min) focusing on: integrating BilimQuest into lesson planning; aligning gamified tasks with Bloom's taxonomy; using analytics dashboards for formative assessment.

Participants also engaged in peer mentoring and reflective practice. Qualitative data from these activities were used to contextualize the quantitative findings and to evaluate the feasibility of integrating the portal into regular teaching practice.

3.6 Ethical considerations

All schools granted formal approval for participation. Ethical approval for the study was obtained from the institutional ethics committee (Approval No. 71). Informed consent was obtained from students, teachers, and parents, with special attention to adapted school participants. The study adhered to international ethical standards for research in education, ensuring confidentiality, voluntary participation, and respect for data privacy.

In summary, the methodology combined theory, design, experimentation, and analytics to provide a comprehensive evaluation of BilimQuest as both a teaching tool and a teacher training model. The use of multiple schools and intact classes supports ecological validity, while the study acknowledges that generalizability beyond similar educational contexts should be approached with caution.

The BilimQuest system is available for academic use at https://web-production-d63f.up.railway.app/.

4 Results

4.1 Functional features of the portal

BilimQuest provides two user modes: teacher and student. The teacher interface includes support materials, methodological guidelines, and an analytics dashboard that visualizes student performance at the class, topic, and individual levels. Learning outcomes are represented both numerically and graphically, with progress mapped to Bloom's taxonomy. The student interface offers personalized learning paths, access to subject modules, gamified tasks, and a reward system (points, stars, badges).

These functional features directly address Research Question 2 (RQ2) by supporting teachers' instructional practices and formative assessment through real-time learning analytics and structured feedback.

Figures 1a–d illustrate the portal's key elements: the home page, teacher login, analytics panel, and individual student progress visualization. These components were designed to ensure accessibility, secure data management, and methodological support for teachers.

4.2 Gamification and Bloom's taxonomy integration

Learning modules (e.g., Information Presentation, Computer Graphics, Binary Systems) were structured through three levels of gamified tasks-knowledge recall, comprehension/application, and higher-order problem solving (analysis, evaluation, creation). Examples include:

- "Build a Bridge" (knowledge level): students link terms and concepts to form a logical structure;
- "Reporter" (comprehension/application): learners interpret and deliver information;
- "Cipher" and "Agent" (analysis/evaluation): tasks simulate information security through encryption and problem-solving;
- "Puzzle" (synthesis/creation): combining learned knowledge into new configurations.

Figures 2a–e show sample task interfaces and the alignment with Bloom's levels. This design ensured that students progressed systematically from lower-order to higher-order cognitive tasks, thereby operationalizing Bloom's taxonomy within a gamified learning environment.

4.3 Student outcomes

To comprehensively assess the impact of the BilimQuest e-learning portal on students' learning outcomes, three components were defined: motivational, content (cognitive), and practical. The motivational component reflects students' interest and persistence in completing gamified tasks; the content component demonstrates mastery of informatics concepts organized according to Bloom's taxonomy; and the practical component highlights students' ability to apply knowledge independently in solving tasks and completing project-based assignments within the portal.

This evaluation framework was explicitly designed to address Research Question 1 (RQ1) by examining whether the integration of gamification aligned with Bloom's taxonomy influences motivational, cognitive, and practical learning outcomes.

For each component, specific criteria and indicators were established to differentiate students' performance across three levels—low, medium, and high. This multi-dimensional framework ensures that the analysis captures not only academic achievement but also motivation, engagement, and practical skills (Serik et al., 2025). The criteria and level indicators are presented in Table 2.

4.4 Baseline (diagnostic stage) results

To analyze baseline equivalence between the experimental and control groups, Pearson's chi-square (χ^2) test was applied. The χ^2 test was selected because the data were categorical and classified into three levels (low, medium, high), and the assumptions of independence of observations and adequate expected frequencies were met.

The following hypotheses were tested:

- H_0 (null hypothesis): the levels of knowledge among students in the experimental and control groups are equal;
- H_1 (alternative hypothesis): the levels of knowledge among students in the experimental and control groups are not equal.

The χ^2 statistic was calculated according to the following formula:

$$\chi^2 = \sum_{i=1}^{k} \frac{(Ef - Tf)^2}{Tf}$$

where E_f represents the empirical frequency and T_f the theoretical frequency.

Using the data from the empirical and theoretical frequency distribution tables, Pearson's chi-square (χ^2) value was calculated according to formula. The results are presented in Table 3.

The calculated χ^2 values for the diagnostic stage were: $\chi^2 = 0.003$ (motivational), $\chi^2 = 0.694$ (content), and $\chi^2 = 1.803$ (practical), with degrees of freedom $\nu = 2$. Since all values were below the critical threshold of 5.99 ($\alpha = 0.05$), the null hypothesis (H_0) was accepted, confirming baseline equivalence between the experimental and control groups prior to the intervention.

4.5 Formative stage results

The formative stage evaluated the impact of BilimQuest after its integration into classroom instruction. This stage incorporated data from final questionnaires and student grades for the implemented course modules. The collected results were entered into Excel and processed using established mathematical and statistical methods. On this basis, conclusions were drawn regarding students' knowledge, skills, and abilities, with particular attention to the development of motivation, content mastery, and practical competencies.

As a result of the formative stage, Pearson's chi-square χ^2 values were obtained for all three components: $\chi^2 = 25.87$ for the motivational component, $\chi^2 = 27.22$ for the content component, and $\chi^2 = 24.70$ for the practical component. All calculated values substantially exceeded the critical value of 9.21 ($\alpha = 0.01$), indicating statistically significant differences between the experimental and control groups in favor of the experimental group (H_1).

The distribution of students' knowledge levels by motivational, content, and practical components in both experimental and control groups is illustrated in Figure 3. The results demonstrate that students in the experimental group achieved markedly higher proportions at medium and high levels across all components, whereas the control group predominantly remained at the low level.

These findings indicate that the null hypothesis (H_0) must be rejected and the alternative hypothesis (H_1) accepted. The results provide empirical evidence that integrating gamification with Bloom's taxonomy through the BilimQuest portal leads to statistically significant improvements in students' motivation, conceptual understanding, and practical skills.

Importantly, the strongest effects were observed in higher-order cognitive outcomes, suggesting that structured gamified tasks can support the development of analysis, evaluation, and creation skills in secondary computer science education. At the same time, the findings

10.3389/feduc.2026.1749909

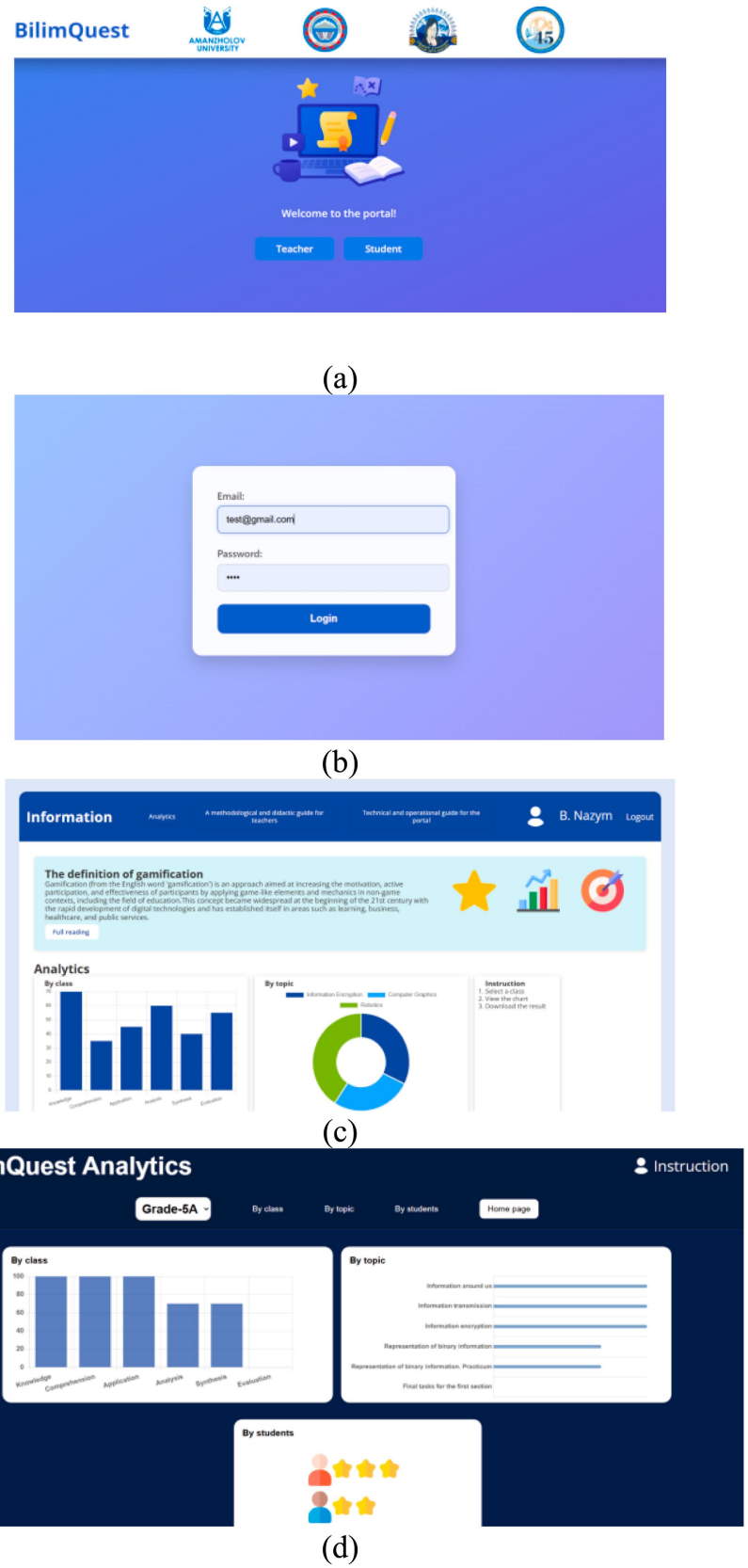
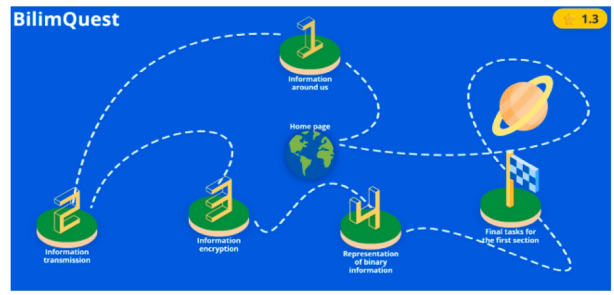


FIGURE 1
BilimQuest portal: **(a)** home page of the electronic learning portal; **(b)** teacher login page; **(c)** teacher interface; **(d)** analytics section.

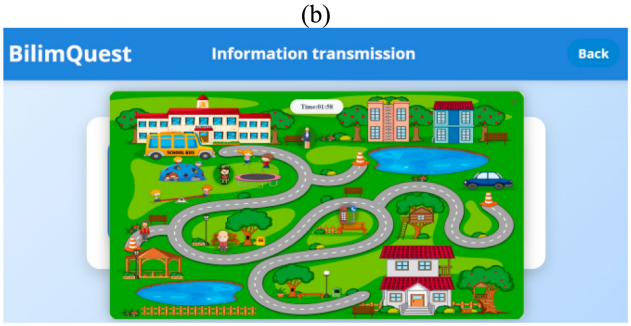
10.3389/feduc.2026.1749909



(a)



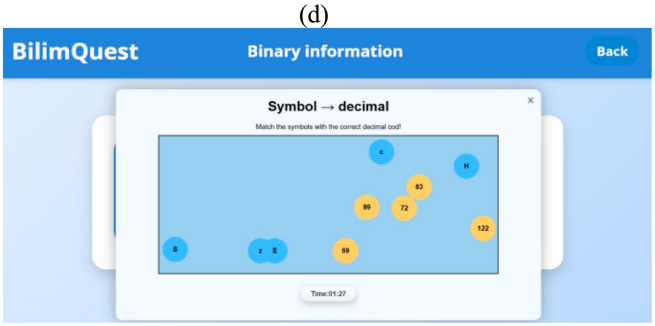
(b)



(c)



(d)



(e)

FIGURE 2
Examples of BilimQuest gamified tasks mapped to Bloom's taxonomy levels: **(a)** "Information Galaxy" thematic module page; **(b)** "Build a Bridge" game for the topic Information Around Us (knowledge level); **(c)** "Reporter" game for the topic Delivering Information (application level); **(d)** "Encryption" game for the topic Encrypting Information (analysis level); **(e)** "Magic Balls" game for the topic Encrypting Information (synthesis/creation level).

TABLE 2 Criteria and level indicators of motivational, content, and practical components for assessing the quality of students' knowledge when integrating the BilimQuest e-learning portal into secondary school education.

Component	Criterion	Low level	Medium level	High level
Motivational	Demonstrates interest and persistence in completing gamified tasks within BilimQuest	Shows weak interest, participates passively, requires constant external encouragement to complete tasks	Demonstrates situational interest, occasionally engages with tasks, needs partial teacher support to stay motivated	Actively engages with gamified activities, demonstrates strong intrinsic motivation, and shows persistence in achieving higher levels
Content (cognitive)	Mastery of informatics concepts and progression through Bloom's taxonomy levels	Knowledge is fragmented, student relies mainly on recall tasks, struggles with comprehension and application	Demonstrates understanding of basic concepts, performs comprehension and application tasks, occasionally attempts higher-order activities	Confidently progresses through Bloom's taxonomy, analyzes, evaluates, and creates digital products, demonstrates critical thinking
Practical	Ability to apply knowledge independently in solving tasks and projects through BilimQuest	Completes only simple tasks with significant errors, requires direct assistance	Successfully performs standard tasks, shows partial independence, applies learned methods with guidance	Independently solves complex tasks, develops creative solutions, applies knowledge to new situations and demonstrates adaptability

TABLE 3 Pearson's χ^2 values at the diagnostic stage.

Component	χ^2	df	Critical value (α = 0.05)	Decision
Motivational	0.003	2	5.99	H_0 accepted
Content	0.694	2	5.99	H_0 accepted
Practical	1.803	2	5.99	H_0 accepted

should be interpreted within the study's context, as the results are based on an embedded mixed-methods design conducted in three schools.

4.6 Qualitative findings supporting the quantitative results

To achieve a clearer interpretation of the findings, qualitative data were collected concurrently with the quantitative phase of the study. This approach made it possible to better understand the underlying reasons for the differences observed between the experimental and control groups.

Teacher observations indicated that the gradual sequencing of learning tasks according to Bloom's taxonomy encouraged students to engage more actively with learning activities rather than focusing solely on task completion. In particular, teachers reported that students in the experimental group showed greater persistence when facing difficulties, increased independence in problem-solving, and more frequent use of analytical thinking during gamified learning situations.

With regard to practical skills, teachers observed that a number of students began applying previously learned concepts to unfamiliar tasks. This observation is consistent with the increase in medium and high performance levels identified in the experimental group during the formative stage. In addition, teachers emphasized that access to real-time learning analytics supported timely formative feedback, which contributed to maintaining student engagement throughout the instructional process. Taken together, the qualitative findings corroborate the quantitative results by illustrating how the pedagogical design of BilimQuest—combining gamification with Bloom's cognitive hierarchy—translated into observable changes in student behavior, motivation, and cognitive engagement.

5 Discussion

The findings demonstrated notable improvements across motivational, content, and practical components, confirming that BilimQuest enhanced both engagement and mastery of cognitive skills. The results of this experiment prove that the integration of gamification and Bloom's taxonomy through the BilimQuest portal significantly influenced students' knowledge levels across three key components-motivational, content, and practical. Improvements were observed not only in sustaining interest and learning motivation, but also in the assimilation of subject content and the ability to apply knowledge in practice, thereby confirming the comprehensive pedagogical impact of the portal.

From a student-centered perspective, the results indicate that structured gamified learning environments can effectively support the development of higher-order thinking skills, which are often challenging to cultivate in secondary computer science education. The observed gains in analysis, evaluation, and creation tasks suggest that aligning game mechanics with Bloom's cognitive hierarchy enables learners to move beyond surface-level engagement toward deeper conceptual understanding and problem-solving competence (Quispe Ccoa et al., 2023; Tangatarova et al., 2025; Tolentino et al., 2025).

These results are consistent with earlier research reporting that gamified platforms such as Kahoot! and Quizizz increase student engagement and participation (Mandai et al., 2024; Wang and Tahir, 2020). However, unlike many existing ICT tools that mainly emphasize recall-level activities, BilimQuest was explicitly structured around

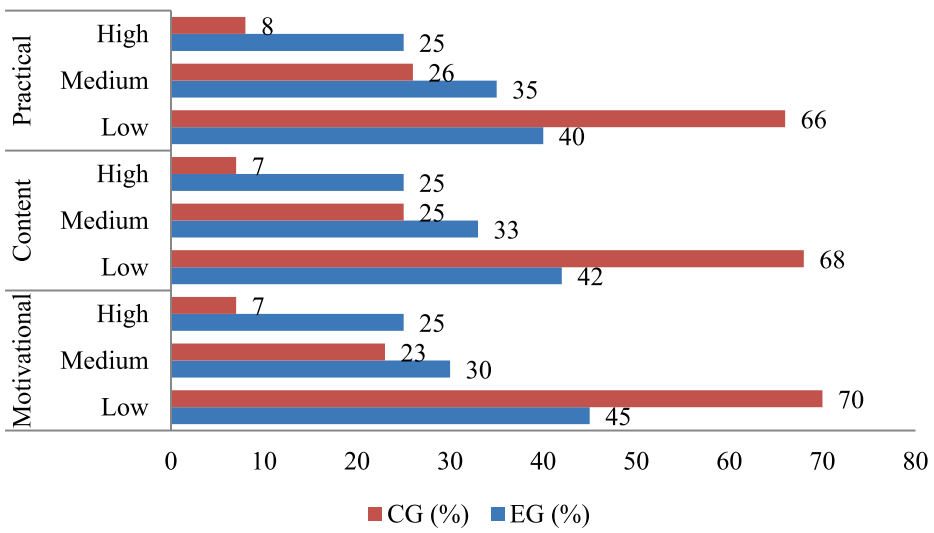


FIGURE 3
Distribution of students' knowledge levels by motivational, content, and practical components.

Bloom's taxonomy, thereby ensuring progression from lower-order to higher-order thinking skills. In this respect, the present study extends prior findings by demonstrating that gamification alone is insufficient unless it is embedded within a cognitively structured pedagogical framework.

The positive outcomes observed in the formative experiment-particularly the improvements in analysis, evaluation, and creation tasks-support the argument that gamification can be an effective pedagogical mechanism when systematically aligned with cognitive objectives. This aligns with the conclusions of Zainuddin et al. (2023) and Ramírez et al. (2024), who emphasized the necessity of scaffolding game mechanics to higher-order competencies. The novelty of BilimQuest lies in embedding Bloom's framework directly into game design, enabling both teachers and students to monitor learning progress through detailed analytics. The key theoretical contribution of this study lies in empirically validating a combined gamification–Bloom's taxonomy model, where cognitive progression is not an implicit by-product of gameplay but an explicitly designed and measurable learning trajectory.

From a teacher-centered perspective, the findings demonstrate that BilimQuest contributes to instructional practice by providing educators with real-time analytics aligned with Bloom's taxonomy, enabling formative assessment and evidence-based instructional decision-making. Teachers can monitor not only task completion but also students' cognitive progression, allowing them to adjust instructional strategies, differentiate support, and identify learners who require targeted intervention. This addresses a critical gap in many gamified systems, which typically lack pedagogically meaningful feedback mechanisms for teachers (Zainuddin et al., 2023; Hamari et al., 2016; Tolentino et al., 2025).

At the same time, certain limitations must be acknowledged. The experiment was confined to three schools in East Kazakhstan and focused on 5th-grade computer science modules, which may restrict the generalizability of the results. Furthermore, while the analytics provided valuable insights into student performance, the system did not yet fully capture qualitative aspects of learning such as collaboration and creativity. Future research should expand the scope to other grade levels and subject areas, integrate the portal with existing learning management systems, and explore its role in fostering transversal competencies such as communication and problem-solving. In addition, an embedded mixed-methods design limits causal inference beyond the studied context, and longer-term effects on knowledge retention were not examined.

Based on these limitations, future research should expand the implementation of BilimQuest to different grade levels, subject domains, and educational contexts, as well as incorporate longitudinal designs to assess sustained learning outcomes. Further development of the platform may include enhanced analytics for collaborative learning and integration with national or institutional learning management systems to support scalability and wider adoption.

Overall, the discussion highlights that BilimQuest not only validates previous findings on gamification's motivational benefits but also extends them by demonstrating how explicit alignment with Bloom's taxonomy can elevate learning outcomes and provide a scientifically grounded methodological framework for ICT-supported education. The study contributes to the literature by offering a replicable design model for cognitively structured gamified learning environments and provides practical guidance for educators seeking to implement competency-based digital instruction in secondary education.

6 Conclusion

This study confirms the pedagogical efficacy of integrating gamification with Bloom's taxonomy through the BilimQuest e-learning portal. This study developed and empirically tested the BilimQuest e-learning portal as an innovative methodological tool that combines gamification with Bloom's taxonomy for secondary computer science education. The results of the pedagogical experiment confirmed that the portal significantly improved students' motivation, knowledge retention, and performance in higher-order cognitive tasks compared to traditional instruction.

The scientific novelty of the study lies in the systematic integration of Bloom's taxonomy into gamified tasks, supported by built-in analytics that allow teachers to monitor student progress across cognitive

levels. By making cognitive progression explicit and observable, the proposed approach advances existing gamification research by demonstrating how motivational game elements can be transformed into structured mechanisms for higher-order learning. This approach demonstrates that gamification, when grounded in a strong pedagogical framework, can move beyond simple engagement mechanisms and foster deeper, structured learning.

The practical significance of BilimQuest is twofold: it provides a scalable digital platform for schools that supports curriculum-based learning objectives, and it contributes to the professional development of future computer science teachers by equipping them with tools to design and implement gamified lessons. For practicing teachers, the platform offers actionable formative assessment data aligned with Bloom's taxonomy, supporting evidence-based instructional decisions and differentiated learning pathways. The portal's architecture, based on modern web technologies, ensures accessibility and adaptability for diverse educational settings.

Nevertheless, the study acknowledges certain limitations related to sample size, geographic focus, and subject scope. Future research should extend testing to other disciplines and educational levels, investigate long-term impacts, and refine the portal's features to capture broader aspects of learner development, including collaborative and transversal competencies. Despite these limitations, the findings provide strong evidence that BilimQuest represents an effective and innovative pathway for integrating gamification and Bloom's taxonomy into modern secondary education. Overall, the research contributes to the interdisciplinary dialogue between pedagogy, ICT engineering, and educational psychology by offering a replicable model for cognitively structured gamified learning environments.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by S. Amanzholov East Kazakhstan University, Ust-Kamenogorsk. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation in this study was provided by the participants' legal guardians/next of kin.

Author contributions

NB: Investigation, Methodology, Resources, Visualization, Writing – review & editing. SA: Funding acquisition, Investigation, Methodology, Project administration, Visualization, Writing – original draft. EP: Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing. RN: Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

Anderson, L. W., and Krathwohl, D. R. (Eds.) (2001). *A taxonomy for learning, teaching, and assessing: a revision of Bloom's taxonomy of educational objectives*. New York, NY: Longman.

Boud, D., and Falchikov, N. (2007). Rethinking assessment in higher education. *Assess. Eval. High. Educ.* 32, 507–522. doi: 10.1080/02602930601095908

Boulden, D. C., Hurt, J. W., and Richardson, M. K. (2021). Gamified formative assessment and student performance: examining Quizizz use in middle school science classrooms. *J. Sci. Educ. Technol.* 30, 355–365.

Damayanti, N., Hartono, Subali, B., Nugroho, S. E., and Sureeporn, K. (2020). Items analysis of physics assessment based on cognitive level of high order thinking skills in Bloom taxonomy. *J. Phys. Conf. Ser.* 1521:022022. doi: 10.1088/1742-6596/1521/2/022022

Davletova, A., Tazhigulova, A., Myrzanova, A., Zhalgasbekova, Z., and Mukhanova, Z. (2025). Motivation through play: gamification as a tool to increase student engagement. *Int. J. Innov. Res. Sci. Stud.* 8, 1853–1861. doi: 10.53894/ijirss.v8i4.8246

Dichev, C., and Dicheva, D. (2017). Gamifying education: what is known, what is believed and what remains uncertain: a critical review. *Int. J. Educ. Technol. High. Educ.* 14, 1–36. doi: 10.1186/s41239-017-0042-5

Dicheva, D., Dichev, C., Agre, G., and Angelova, G. (2015). Gamification in education: a systematic mapping study. *Educ. Technol. Soc.* 18, 75–88.

Dong, L., Jiang, S., Xu, F., Duan, S., and Hsieh, C.-J. (2025). From simplification to sophistication: secondary students' conceptual change in understanding machine learning model decision making. *Br. J. Educ. Technol.* 57:423. doi: 10.1111/bjet.70024

Erkek, G. (2022). Examination of text-based questions in secondary school Turkish textbooks according to the renovated Bloom taxonomy. *Proceedings of the international conference on research in education and science (ICRES)*, 8, 32–41.

Fernandes-Henriques, P., and Carvalho, A. A. (2021). Classcraft: promoting students' motivation through gamification. *Educ. Inf. Technol.* 26, 3279–3298.

Forehand, M. (2010). Bloom's taxonomy Emerging perspectives on learning, teaching, and technology. Vanderbilt University, Nashville, TN. 41, 47–56.

Grover, S., and Pea, R. (2013). Computational thinking in K–12. *Educ. Res.* 42, 38–43. doi: 10.3102/0013189X12463051

Hamari, J., Shernoff, D. J., Rowe, E., Coller, B., Asbell-Clarke, J., and Edwards, T. (2016). Challenging games help students learn. *Comput. Hum. Behav.* 54, 170–179. doi: 10.1016/j.chb.2015.07.045

Ke, F. (2016). Designing and integrating purposeful learning in game play: a systematic review. *Educ. Technol. Res. Dev.* 64, 219–244. doi: 10.1007/s11423-015-9418-1

Kgosietsile, R. (2023). Mapping game-based assessments to Bloom's taxonomy: a framework for progressive cognitive development. *Int. J. Educ. Technol. Learn.* 18, 45–56.

Krathwohl, D. R. (2002). A revision of Bloom's taxonomy: an overview. *Theor. Pract.* 41, 212–218. doi: 10.1207/s15430421tip4104_2

Licorish, S. A., Owen, H. E., Daniel, B., and George, J. L. (2018). Students' perception of Kahoot!'s influence on teaching and learning. *Res. Pract. Technol. Enhanc. Learn.* 13, 1–23. doi: 10.1186/s41039-018-0078-8

Mandai, K., Tan, M. J. H., Padhi, S., and Pang, K. T. (2024). A cross-era discourse on ChatGPT's influence in higher education through the lens of John Dewey and Benjamin Bloom. *Educ. Sci.* 14:614. doi: 10.3390/educsci14060614

Ministry of Education and Science of the Republic of Kazakhstan (2020). *State program for the Development of Education and Science of the Republic of Kazakhstan for 2020–2025*. Astana, Kazakhstan: Ministry of Education and Science of the Republic of Kazakhstan.

Nawaz, I., Cuenen, A., Wets, G., Paul, R., and Janssens, D. (2025). Advancing online road safety education: a gamified approach for secondary school students in Belgium. *Appl. Sci.* 15:8557. doi: 10.3390/app15158557

Nebel, S., Schneider, S., and Rey, G. D. (2016). Mining learning and crafting scientific experiments: a literature review on the use of Minecraft in education and research. *Educ. Technol. Soc.* 19, 355–366.

Plump, C. M., and LaRosa, J. (2017). Using Kahoot! In the classroom to create engagement and active learning. *Manag. Teach. Rev.* 2, 151–158. doi: 10.1177/2379298116689783

Quispe Ccoa, N. M., Farfan Choquehuanca, M. E., and Rucano Paucar, F. H. (2023). An application of the Quizizz gamification tool to improve motivation in the evaluation of elementary school students. *Int. J. Inf. Educ. Technol.* 13, 544–550. doi: 10.18178/ijiet.2023.13.3.1837

Ramírez, D., Torres, A., and González, M. (2024). Gamification and school engagement: a systematic review through the lens of Bloom's taxonomy. *J. Comput. Assist. Learn.* 40, 88–104.

Resnick, M., Maloney, J., Monroy-Hernández, A., Rusk, N., Eastmond, E., Brennan, K., et al. (2017). Scratch: programming for all. *Commun. ACM* 52, 60–67. doi: 10.1145/1592761.1592779

Ritter, F., and Schlomske-Bodenstein, N. (2025). Fostering algorithmic thinking within a productive-failure-based workshop utilizing AI. *Proceedings of the IEEE Global Engineering Education Conference (EDUCON)*, 1–6.

Sailer, M., Hense, J. U., Mayr, S. K., and Mandl, H. (2017). How gamification motivates: an experimental study of the effects of specific game design elements on psychological need satisfaction. *Comput. Educ.* 106, 71–88. doi: 10.1016/j.compedu.2016.12.004

Sailer, M., and Homner, L. (2020). The gamification of learning: a meta-analysis. *Educ. Psychol. Rev.* 32, 77–112. doi: 10.1007/s10648-019-09498-w

Serik, M., Tleuzhanova, K., and Nurgaliyeva, S. (2025). Enhancing master's-level STEM education through AI-driven IoT projects: a Kazakhstan experiment. *Int. J. Inf. Educ. Technol.* 15, 812–820. doi: 10.18178/ijiet.2025.15.10.2407

Shaukat, M. R., Kukreti, M., Jain, V., Aziz, M. F., and Sakiti, M. K. A. (2025). Exploring the dynamics of playing for knowledge. *Int. J. Innov. Res. Sci. Stud.* 8, 1745–1758. doi: 10.53894/ijirss.v8i3.6882

Singh, K. D., and Singh, P. D. (2025). Edge-AI driven gamification in engineering education for improving student engagement. *Comput. Appl. Eng. Educ.* 33:e70076. doi: 10.1002/cae.70076

Southgate, E., Smith, S. P., and Cheers, H. (2020). Immersive virtual reality, gamification, and creativity. *J. Educ. Comput. Res.* 58, 566–590. doi: 10.1177/0735633119854022

Stoecklin, D. (2020). Interactive content with H5P: a gamification tool for higher education. *J. Appl. Res. High. Educ.* 12, 987–999.

Tangatarova, Z., Nurgalym, K., Nishanbayeva, S., Sadirbekova, D., Kozhamkulova, N., and Akimbekova, S. (2025). Gamification's effect on the acquisition of patriotic behaviors in a primary school social studies course. *Int. J. Inf. Educ. Technol.* 15, 1607–1615.

Tolentino, A. P., Santos, E. S., da Silva, S. E. G., de Araújo, H. X., and Rodrigues, D. C. (2025). Digital gamification: benefits and challenges in education. *Int. J. Inf. Educ. Technol.* 15, 902–911. doi: 10.18178/ijiet.2025.15.5.2296

Wang, A. I., and Tahir, R. (2020). The effect of using Kahoot! For learning: a literature review. *Comput. Educ.* 149:103818. doi: 10.1016/j.compedu.2020.103818

Xu, B., and Wang, M. (2021). Adaptive learning systems in mathematics education: a systematic review. *Comput. Educ.* 168:104193. doi: 10.1016/j.compedu.2021.104193

Zainuddin, Z., Alba, A., Gunawan, T., Armanda, D., and Zahara, A. (2023). Implementation of gamification and Bloom's digital taxonomy-based assessment: a scale development study with mixed-methods sequential exploratory design. *Interact. Technol. Smart Educ.* 20, 512–533. doi: 10.1108/ITSE-02-2022-0029